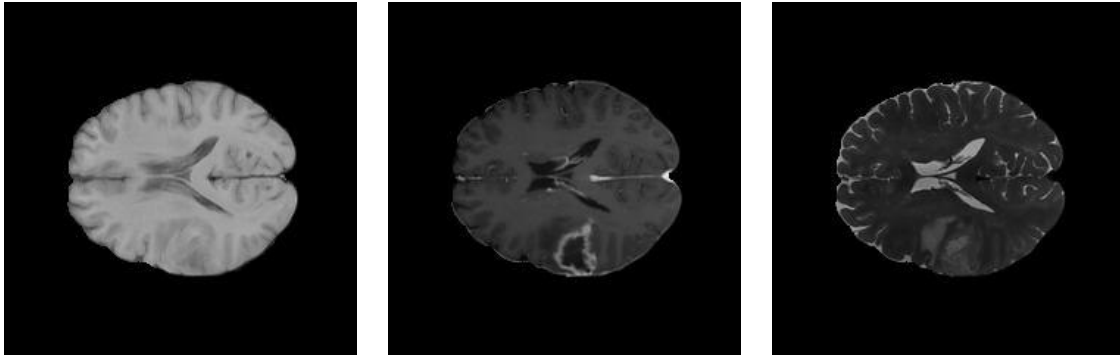


Tumour Detection and Segmentation from Multi-Sequence 2D Brain MRI Slices (UCSF-PDGM)



Problem

1. Classification:
 - Detections of tumour in each slice using `is_tumorous`
2. Segmentation:
 - Pixel localisation using `tumor_mask`.

Data Structure

Dataset

Field	Description
<code>volume_id</code>	Unique patient identifier
<code>slice_id</code>	Index of slice of MRI (155 slices)
<code>t1</code>	2D Pre-contrast T1 weighted MRI slice
<code>t1c</code>	2D Post-contrast T1 weighted MRI slice
<code>t2</code>	2D T2 weighted MRI slice
<code>tumor_mask</code>	2D tumor segmentation mask
<code>is_tumorous</code>	Whether any tumour label is present on the slice
<code>tumor_type</code>	Final diagnosis
<code>who_grade</code>	WHO tumour grade
<code>sex</code>	Patient sex M or F

Field	Description
age	Patient age at MRI

Tumour mask

Value	Label	Description
0	Background	No tumour
1	NCR/NET	Necrotic and non-enhancing tumour core
2	ED	Peritumoral edema / surrounding FLAIR abnormality
4	ET	GD-enhancing tumour

Tumour Grades

Tumour Region of Interest	Labels
No Tumour	0
Whole Tumour	1, 2, 4
Tumour Core	1, 4
Enhancing Tumour	4

All patients have a confirmed diagnosis, after removing missing values there is 414 patients x 155 slices = 64170 slices. `is_tumorous` varies per slice with 38,715 no tumour slices and 25,455 tumour slices.

UCSF_PDGM Dataset - Hugging Face

Baseline

Logistic regression on flattened pixels (`t1`, `t1c`, `t2`) and patient metadata (`sex`, `age`) to predict `is_tumorous`.

Metric

- Classification: AUC (Area Under Curve)
- Segmentation: IOU (Intersection Over Union)

Risk

The primary risk is data leakage at the patient level as each patient has 155 slices, therefore the data must be split by `volume_id` not `slice_id`. `tumor_type` and `who_grade` will be excluded from the model input features as these are consistent across whole patient information rather than on the individual slices.

Team

Individual.

AI Tools

I will try use a combination of LLM models including the Ollama qwen3:1.7b and trialing Claude Sonnet 5 depending on the depth of questions or errors encountered.